OUR WORLD IN DATA • RENEWABLE ENERGY

Global Renewable Energy in 2025

How far have we come across primary energy, electricity generation, and the technologies reshaping the power system?

Sources: Our World in Data; Ember (2026); Energy Institute Statistical Review of World Energy (2025).

2025

WHY THE TRANSITION REMAINS URGENT

75% of global GHG emissions still come from fossil fuels

¾

of global greenhouse gas emissions come from burning fossil fuels for energy.

5M+

premature deaths each year are linked to local air pollution from fossil fuel combustion.

2

low-carbon pillars—renewables and nuclear—are needed to reduce CO_2 and air pollution.

CO_2

Industrial-era lock-in

Since the Industrial Revolution, most national energy systems have become dominated by fossil fuels.

PM

Climate and health costs

The same combustion driving emissions also produces air pollution with severe mortality impacts.

↗

Transition imperative

Modern renewables exclude traditional biomass and include solar, wind, hydro, geothermal, bioenergy, wave, and tidal.

PRIMARY ENERGY • 2024

Renewables now supply ~14% of global primary energy

~1/7

Primary energy includes electricity, transport, and heating. Because transport and heating remain harder to decarbonize, the renewable share is lower than in electricity alone.

> **Method note:** global primary energy share uses the substitution method referenced in the source.

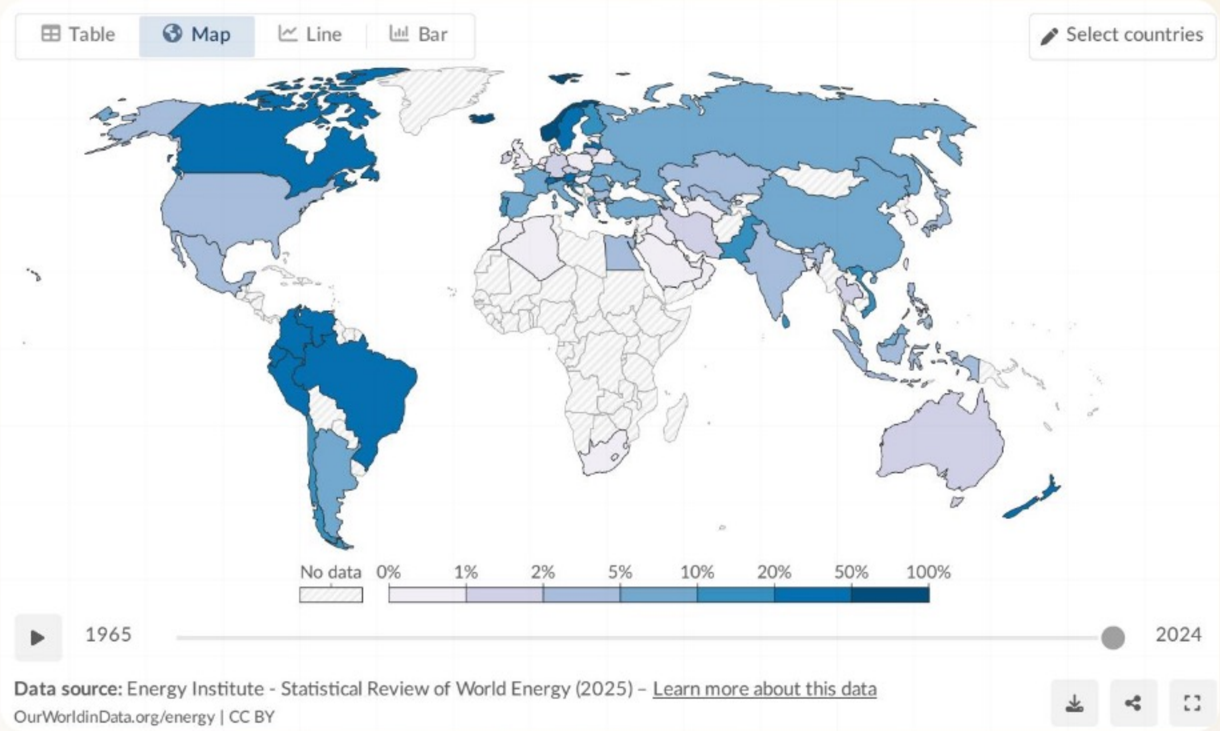

Figure: Share of primary energy from renewables by country, 2024.

RENEWABLE ELECTRICITY GENERATION • 2024

Hydro leads, but wind and solar together nearly match its output

~9,600

TWh of renewable electricity generation across hydro, wind, solar, and other renewables in the brief's technology table.

> Wind and solar accelerated sharply after ~2010, driven by falling costs and policy support, while hydropower remains the largest single renewable source.

ELECTRICITY MIX • 2025

Almost one-third of global electricity now comes from renewables

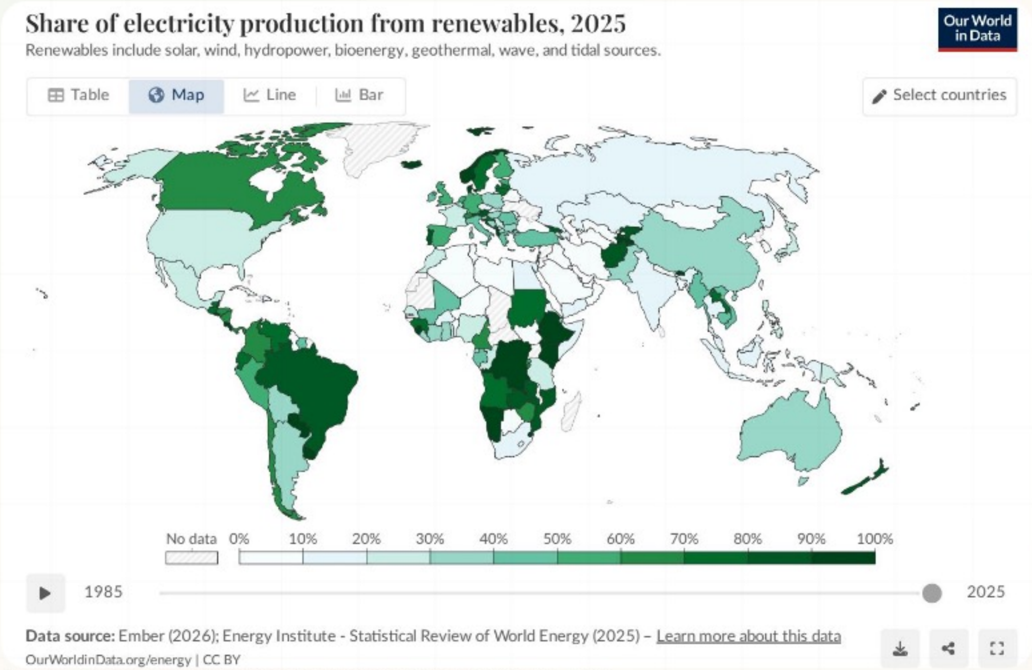

Figure: Share of electricity from renewables by country, 2025.

COUNTRY / REGION	RENEWABLE ELECTRICITY	RELATIVE LEVEL
Norway	~98%	
Brazil	~85%	
Canada	~65%	
China	~35%	
India	~25%	
Saudi Arabia	<2%	

World average is ~30%; several African and Middle Eastern countries remain below 10%.

HYDROPOWER • 2025

Hydropower remains the largest renewable source at ~50% of generation

~4,500

TWh generated globally, roughly half of all renewable electricity.

> Top producers include China, Brazil, and Canada. Hydropower has generated electricity at scale for more than a century.

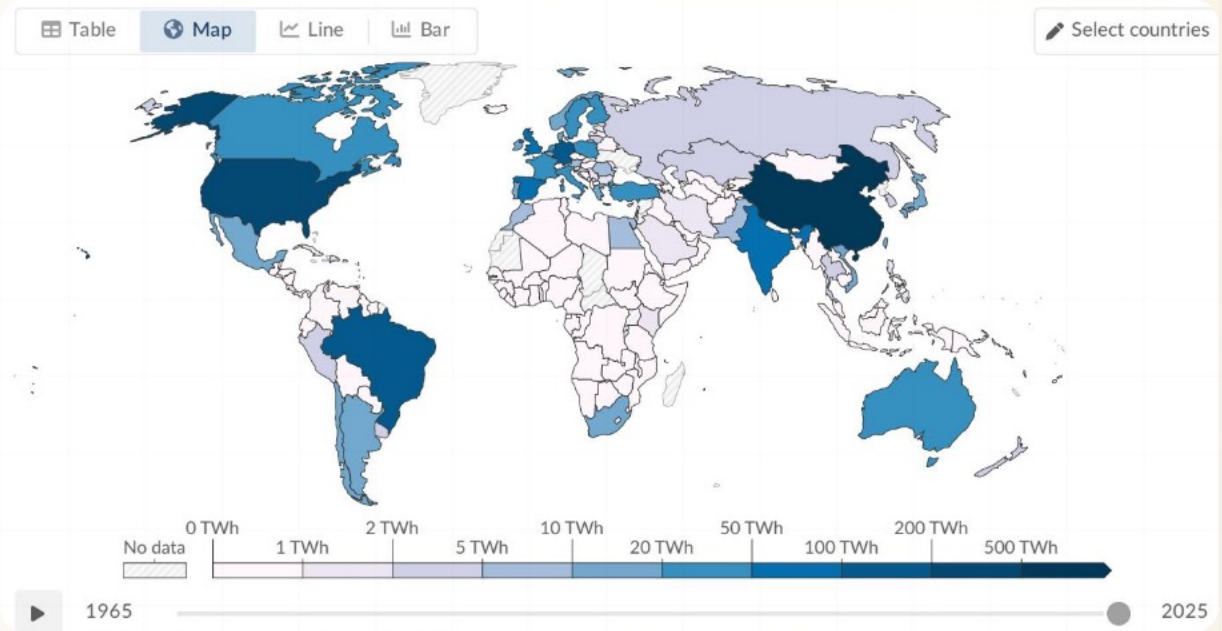

Figure: Hydropower generation by country, 2025.

FASTEST-GROWING TECHNOLOGIES

Wind and solar are catching hydropower on generation—not capacity

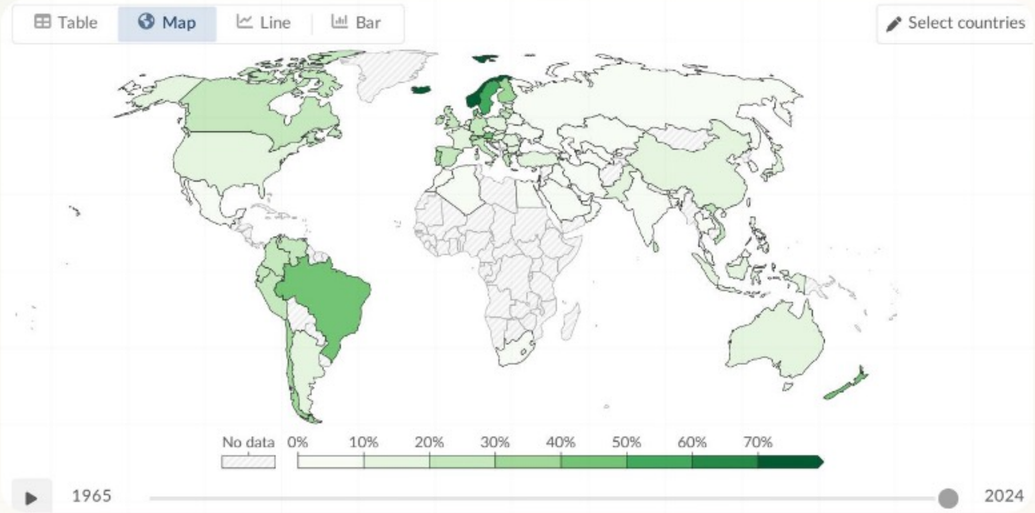

Figure: Wind power generation by country, 2025.

WIND

~2,400

TWh, about ~25% of renewable electricity. Major contributor in Europe, China, and the United States.

SOLAR

~2,000

TWh, about ~21% of renewable electricity. Fastest-growing source, with dramatic growth since 2010.

Combined wind and solar output is ~4,400 TWh—nearly matching hydropower's ~4,500 TWh.

REGIONAL DISPARITIES

Scandinavia and Latin America lead, while many regions remain below 10%

65–98%

Renewable electricity shares in Canada, Brazil, and Norway illustrate high-renewable systems anchored by hydropower.

<10%

Many African and Middle Eastern countries remain below this level, reflecting fossil dependence and limited renewable infrastructure.

> Denmark and Germany show how wind and solar investment can complement hydropower-led transition pathways.

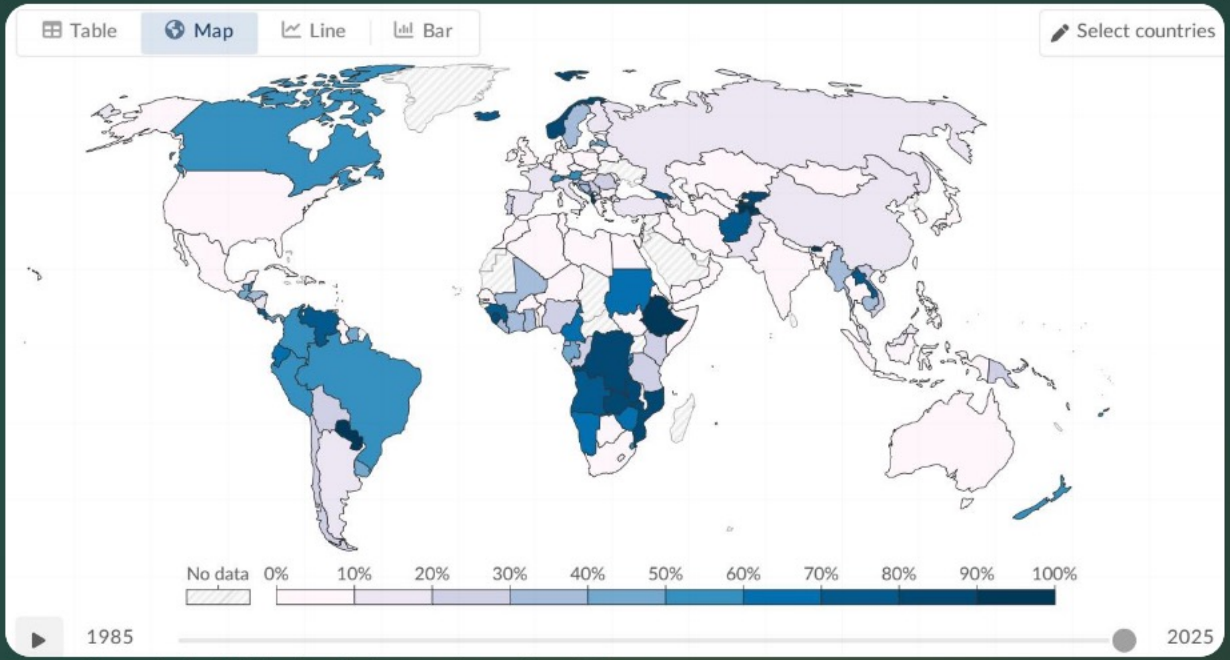

Figure: Share of electricity from hydropower by country, 2025.

POLICY AND STRATEGY IMPLICATIONS

Renewables are growing fast, but the transition must accelerate

~14%
of global primary energy is renewable, showing total energy transition remains early-stage.

~30%
of global electricity is renewable, where the transition has advanced fastest.

Hydro
still provides roughly half of renewable generation, led by China, Brazil, and Canada.

Wind + solar
now nearly match hydro output and are the fastest-growing energy technologies.

Gaps
persist across regions, and transport and heating lag far behind electricity.

Forward agenda: accelerate deployment, grid integration, and policy support where renewable infrastructure is still thin.

Sources: Our World in Data; Ember (2026); Energy Institute Statistical Review of World Energy (2025).